



Detection of bee-associated pathogens in different developmental stages within the nest of the invasive hornet *Vespa velutina nigrithorax* Buysson, 1905

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Abstract The yellow-legged hornet (*Vespa velutina nigrithorax*) is an invasive species in Europe that has created serious environmental issues by feeding on insect pollinators, and it is a major cause of concern for beekeepers due to its predatory activity against *Apis mellifera* colonies. Predation can play a role in transmitting pathogens between bees and hornets, thereby contributing to the spread of diseases in the environment. This study assessed the prevalence and abundance of honey bee- and bumblebee-associated pathogens in different developmental stages of *V. v. nigrithorax* individuals collected from nests in the Aveiro and Coimbra regions of Portugal.

A total of 534 samples (larvae, pupae, adults, and queens) within 38 nests were analysed using qPCR for the main bee pathogens. The most prevalent and abundant pathogen was DWV (70.41%; 4.66×10^{10}), followed by *Nosema ceranae* (42.70%; 6.17×10^6), BQCV (44.76%; 1.64×10^5) and CBPV (33.90%; 2.17×10^9). DWV and other pathogens were detected across all developmental stages, indicating widespread pathogen detection and suggesting multiple exposure routes within the nest with possible vertical transmission. Pathogen loads were significantly higher in the Aveiro region, possibly due to environmental factors and nest maturity. These findings confirm that *V. v. nigrithorax* harbours multiple replicative bee-associated pathogens, raising concerns about interspecific transmission.

Soraia Sousa Santos and Rossella Tiritelli contributed equally to this work.

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Introduction

Biological invasions have been considered a major concern worldwide due to their impact on ecosystems and for generating socio-economic concerns (Hochberg and Gotelli 2005; De Jong and Lester 2023; Chapman et al. 2023). The yellow-legged hornet (*Vespa velutina nigrithorax* Buysson, 1905) is naturally distributed in Southeast Asia, but it was

accidentally introduced in Europe in 2004, registered in France for the first time and has rapidly spread to other countries (Castro and Pagola-Carte 2010; Rortais et al. 2010; Grosso-Silva and Maia 2012; Walter et al. 2024). In 2010, the first individuals were observed in northeastern Spain and the following year, in northwestern Portugal (Castro and Pagola-Carte 2010; Grosso-Silva and Maia 2012). In fact, the expansion in Portugal essentially occurred from the north to the south along the coast, reaching the Coimbra region by 2015 (<https://projects.inia.pt/gesvespa>).

The successful invasion of *V. v. nigrithorax* can be attributed to their adaptation, reproductivity and dispersal strategy (Rortais et al. 2010; Beggs et al. 2011; Requier et al. 2024). As a eusocial insect, the yellow-legged hornet lives in large colonies divided into castes (queen, female workers and males) (Pérez-De-Heredia et al. 2017). After winter diapause, a foundress (fertilized) queen of *V. v. nigrithorax*, establishes its colony by building a primary nest, in which female workers are produced; during the summer season, the colony will build a secondary nest in which it will develop. At the end of the summer, new queens are produced and fertilized by the males before hibernating over winter, while the rest of the colony dies (Rome et al. 2015; Keeling et al. 2017).

Through the different responsibilities of the workers, feeding the larvae is crucial for colony development. The diet of the larvae is mainly composed of protein obtained from insects (Monceau et al. 2013). Among these insects, there is a preference for honey bees, especially for *Apis mellifera*, which accounts for more than 70% of the hornet diet in urbanized areas (Tan et al. 2007; Villemant et al. 2011). Within the honey bee colonies, there are high circulation of several pathogens affecting the individual and colony health (Genersch 2010; Cornman et al. 2012; Steinhauer et al. 2018; Beaurepaire et al. 2020). These pathogens comprise a huge variety of microorganisms, including bacteria, fungi, protozoa, and viruses, with different epidemiology and transmission routes (Higes et al. 2006; Brutscher et al. 2016; Grozinger and Flenniken 2019; Lannutti et al. 2022; Tiritelli et al. 2025).

The predator behaviour of *V. v. nigrithorax* on *A. mellifera* can lead to the transmission of several of these pathogens (Dalmon et al. 2019; Gabín-García

et al. 2021; Rodríguez-Flores et al. 2022, 2024). A few studies proved that some of the bee pathogens are present in the yellow-legged hornet, including, *Nosema ceranae*, *Lotmaria passim*, Kashmir bee virus (KBV), black queen cell virus (BQCV), and deformed wing virus (DWV) (Mazzei et al. 2019; Bartolomé et al. 2024; Cilia et al. 2024a), but there's still a lack of information, i.e., Mazzei et al. (2019) only analysed adult female, Bartolomé et al. (2024) did not specified the sex of the samples and Cilia et al. (2024a, b) only had sampling from one nest.

These pathogens may reduce the lifespan of the yellow-legged hornet and this invasive species could act as a spillback reservoir for native bees, making it essential to understand their interspecific transmission. This study aimed to analyse the presence, abundance and distribution of the main bee- and bumblebee-associated pathogens in different development stages (larvae, pupae and adults) of *V. v. nigrithorax*. The pathogen panel targeted the most prevalent viral (DWV, BQCV, CBPV, SBV, ABPV), trypanosomatids (*L. passim*, *Crithidia mellificae*, and *C. bombi*), neogregarine (*Apycistis bombi*) and microsporidian (*N. ceranae*) pathogens currently affecting managed and wild bees in Europe. Although additional pathogens, such as *Nosema apis* or Israeli Acute Paralysis Virus could be considered, these are currently rare or sporadic in the study region and were therefore considered of lower epidemiological relevance.

Material and methods

Sampling and collection site

Nests were collected from the Aveiro and Coimbra regions, located in northern and central Portugal, respectively. A total of 38 nests were sampled, with 13 from Aveiro and 25 from Coimbra (Fig. 1 and Table S1). Active nest's locations were provided by local authorities', and only untreated nests up to 10 m were collected. Nests were manually collected directly by field technicians wearing protective suits and gloves, between 30 July and 21 November 2024. Nests were transported to the laboratory in insulated containers under cold conditions and subsequently stored at -20°C until processing. The number of individuals selected for each developmental stage

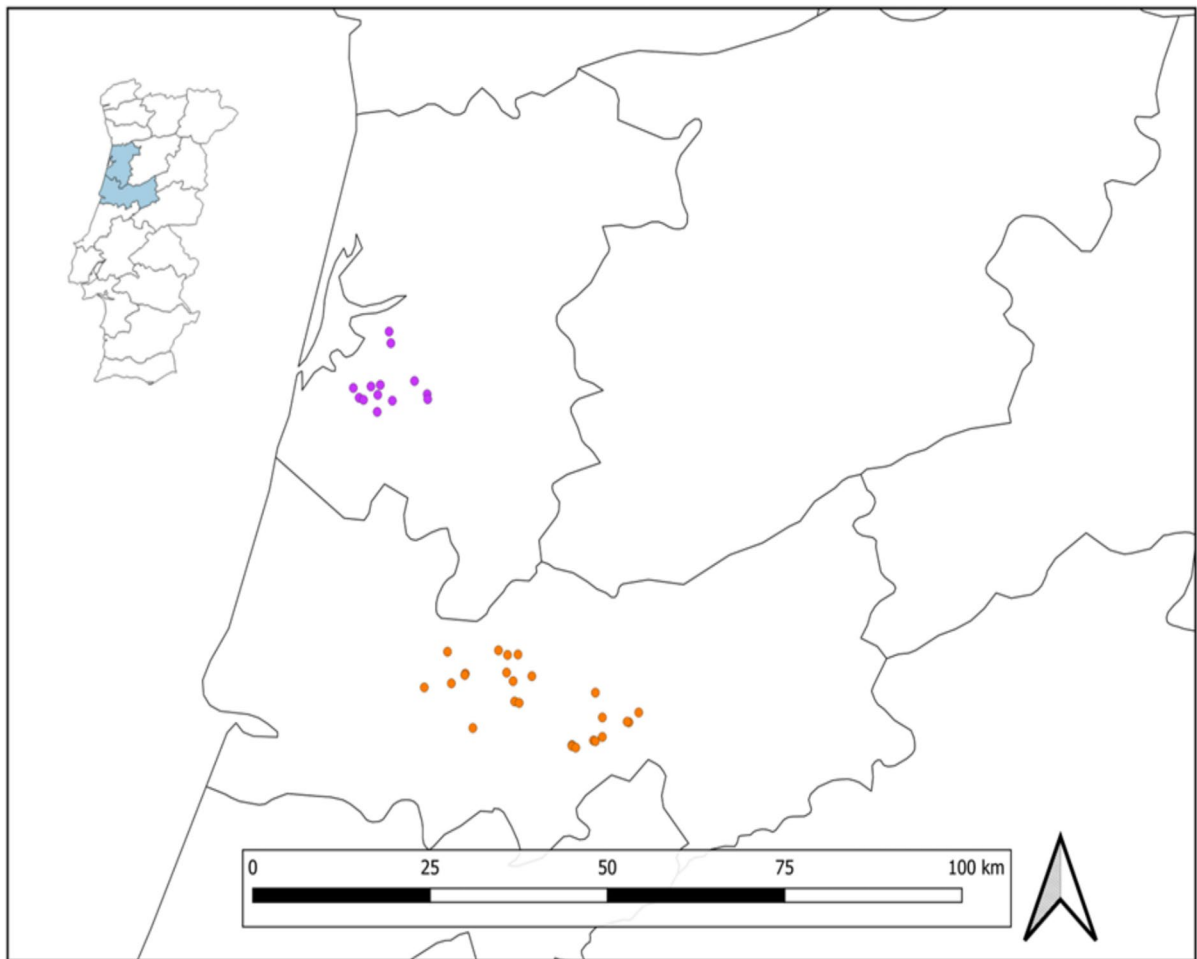


Fig. 1 Political map of Portugal, showing the geographical localization of the collected *V. v. nigrithorax* nests in the Coimbra (orange dots) and Aveiro (purple dots) regions

varied according to availability within the nest; however, whenever possible, 30 individuals per stage were sampled (Table S1). Three developmental stages were considered: adult workers, pupae, and larvae. Queens, when present, were also included in the analysis. None of the samples exhibited symptoms of pathogen occurrence. The sex of adult individuals was determined based on genital morphology. All samples were stored at -80°C until further analysis.

Pathogens' detection

For each nest, pools of 5 individuals were created, if possible, for each stage investigated (Table S1). Individuals from each pool were washed in 95%

ethanol before extraction to eliminate any external microbiological contamination. Pool homogenization and DNA and RNA extraction were performed using a Quick DNA Microprep Plus Kit and a Quick RNA Microprep Plus Kit (Zymo Research, Irvine, CA, USA), as previously reported (Cilia et al. 2024a). The extracted nucleic acids were eluted in 200 μl of DNAase-RNase-free water and kept at -80°C until the qPCR analysis.

A quantitative Real-Time PCR (qPCR) analysis was performed to determine the abundance of each pathogen in the samples using the extracted DNA (for *N. ceranae*, *L. passim*, *Crithidia mellificae*, *C. bombi* and *Apicystis bombi*) and RNA (for viruses). The primers used for the qPCRs are reported in Table S2.

The qPCR was performed in a total reaction volume of 10 µl for each target gene using SYBRTM green assays as previously reported (Cilia et al. 2024b; Tiritelli et al. 2024a). The SYBR PowerUpTM SYBRTM Green Master Mix (ThermoFisher Scientific, Waltham, MA, USA) and the Power SYBRTM RNA-to-CTTM (1-Step) Kit (ThermoFisher Scientific) were used for the DNA and RNA, respectively. The qPCRs were carried out using a QuantStudioTM 3 Real-Time PCR System (ThermoFisher Scientific), according to the protocols for each gene sequence (Chantawannakul et al. 2006; Mazzei et al. 2018; Cilia et al. 2018; Fernández et al. 2020; Buendía-Abad et al. 2023).

As a positive control for each pathogen, DNA and RNA previously extracted from positive honey bees were used, while double-distilled sterile water was used as a negative control. All the analyses were carried out in triplicate. A standard curve from 1×10^1 to 1×10^9 copies was created for each target gene, as previously reported (Cilia et al. 2022a, 2023), using the amplification and quantification protocols (Chantawannakul et al. 2006; Mazzei et al. 2018; Cilia et al. 2018; Fernández et al. 2020; Buendía-Abad et al. 2023).

Strand-specific RT-qPCR

The replication forms of the detected viruses (DWV, CBPV, ABPV, BQCV and SBV) in all *V. velutina* pools were evaluated by the specific identification of negative-strand viral RNA using strand-specific RT-PCRs, as previously reported (Mazzei et al. 2019; Cilia et al. 2022b; Rodríguez-Flores et al. 2024). Briefly, a two-step RT-PCR was used for the specific detection of each pathogen, using Superscript III reverse transcriptase (Invitrogen, Waltham, MA, USA) in a reaction volume of 20 µl, following the manufacturer's instructions. Further, 5 µl of cDNA were used as a template for the PCR reaction, which was carried out with DreamTaqTM Hot Start DNA Polymerase (ThermoFisher Scientific) using the specific primer pairs and thermal cycling for each pathogen (Cilia et al. 2022a). Amplicons were visualised on a 2% agarose gel. Samples in which double-strands were visualized in the agarose gel were considered replicative.

Statistical analysis

Exploratory analyses were performed by calculating abundance and prevalence parameters both overall and in relation to developmental stages. Prevalence was calculated over the total number of samples collected, while abundance was calculated as the mean copy number obtained from two technical qPCR replicates of the same biological sample, consisting of a pool of individuals. The averaged values were then log₁₀-transformed prior to statistical analyses to normalize the data distribution and allow comparisons across samples and pathogens.

To test the probability of infection, a binomial GLM model was fitted, with prevalence values as the dependent variable and pathogens, developmental stages (larva, pupa, adult), and municipalities (Aveiro, Coimbra, Lousã, Miranda do Corvo) as independent variables. These independent variables were also tested on abundance values by fitting a glmmTMB model. Data related to queens were excluded from these analyses. Following model fitting, an ANOVA was performed. Pairwise comparisons between groups were conducted using a post hoc Tukey test to assess statistical differences. The optimal model was selected using the Akaike Information Criterion (AIC). The same models, using prevalence and abundance values, were fitted to the samples obtained from nests where a queen was present. In this case, pathogens, developmental stages (larva, pupa, adult, queen) and ID. Nest were used as an independent variable. Seasonal differences in infection were tested using models in which the prevalence and abundance of each pathogen were treated as dependent variables, and month and municipality were included as explanatory variables. In this analysis, the pathogens *L. passim* and *C. bombi* were excluded due to the very low number of positive samples (26 and 25, respectively).

The significant threshold was set at 0.05. All the analyses were conducted in R 4.4.2 (r-project.org). Data manipulation, analysis, and graphical representation were performed using the *car*, *glmmTMB*, *ggplot2*, *ggpubr*, *emmeans*, *logistf* (Wickham 2009; Brooks et al. 2017; Champely et al. 2022; Fox 2022; Kassambara 2023; Lenth 2024; Heinze et al. 2025).

Results

A total of 534 samples were analysed, covering different development stages: larvae ($N = 175$), pupae ($N = 173$), adults ($N = 183$), and queens ($N = 3$). All samples tested negative for KBV and *C. mellifica*. Among the pathogens investigated, DWV was the most prevalent (70.41%), followed by BQCV (44.76%) and *N. ceranae* (42.70%) (Table 1). DWV was also the most abundant pathogen ($4.66^{10} \pm 5.75^{11}$), followed by CBPV ($2.17^9 \pm 3.66^{10}$) and SBV ($1.41^7 \pm 2.23^8$) (Table 1). All 9 pathogens were detected across the three developmental stages (larva, pupa and adult). However, among the three queens sampled, only DWV, CBPV, BQCV, SBV, and *C. bombi* tested positive (Fig. 2). In all samples that scored positive to investigated viruses, they were found replicative.

The model that included the three developmental stages (larva, pupa, and adult) identified a significant effect of pathogens ($\chi^2 = 1191.04$, $df = 8$, $P < 0.001$) and municipality ($\chi^2 = 32.82$, $df = 3$, $P < 0.001$) on prevalence values (Fig. 3a). The model explained 21.67% of the overall variance. Similarly, the abundance model showed a significant effect on the municipality ($\chi^2 = 39.53$, $df = 3$, $P < 0.001$) and pathogens ($\chi^2 = 1079.58$, $df = 3$, $P < 0.001$). However, the developmental stage was not a significant factor in either prevalence or abundance models. Post-hoc comparisons between municipalities revealed significantly higher pathogen prevalence in Aveiro compared to Coimbra and Lousã. Likewise, abundance values were significantly higher in Aveiro

than in the other three municipalities. In post-hoc comparisons among pathogens, DWV exhibited significantly higher prevalence and abundance than all other pathogens in both models (Fig. 3b). The complete results are available in Table S3.

Models that included data on nests where queens were sampled found no statistically significant differences between developmental stages (Fig. 4a and Fig. 4b). However, significant differences were observed among pathogens for both prevalence ($\chi^2 = 230.31$, $df = 8$, $P < 0.001$) and abundance models ($\chi^2 = 223.34$, $df = 8$, $P < 0.001$). The prevalence model explained 43.31% of the overall variance. In the prevalence model, DWV was significantly more likely to be detected than the other pathogens (Fig. 4c). Besides, the pathogen abundance was influenced by the sampled nest, as demonstrated by the significantly low loads of nest 23 compared to nest 30 ($\chi^2 = 7.460$, $df = 2$, $P < 0.05$) (Fig. 4d). Table S4 shows the complete results.

The different abundance of pathogens is also evident from their spatial distribution, as their abundance manifests itself with different intensities in the nests sampled in the different Portuguese regions (Fig. 5).

Significant seasonal differences were observed in pathogen prevalence and abundance across months (Fig. 6). In particular, DWV was detected at significantly higher levels in November compared with the other months, BQCV in November and October, CBPV in October compared with September, ABPV in August, and SBV in September (Figure S1 and Table S5). Among municipalities, Aveiro showed a

Table 1 Pathogen prevalence and mean copy number ($\pm$ standard deviation)

Pathogen	N° of positive samples	Prevalence	Mean copy number	Standard deviation
DWV	376	70.41%	4.66×10^{10}	5.75×10^{11}
CBPV	181	33.90%	2.17×10^9	3.66×10^{10}
ABPV	58	10.86%	7.11×10^5	1.64×10^7
BQCV	239	44.76%	1.64×10^5	1.45×10^6
SBV	154	28.84%	1.41×10^7	2.23×10^8
KBV	0	–	–	–
<i>A. bombi</i>	31	5.81%	4.10×10^4	4.32×10^5
<i>N. ceranae</i>	228	42.70%	6.17×10^4	1.39×10^8
<i>L. passim</i>	26	4.87%	1.38×10^5	2.93×10^6
<i>C. mellifica</i>	0	–	–	–
<i>C. bombi</i>	26	4.87%	5.83×10^4	1.06×10^6

DWV, deformed wing virus; CBPV, chronic bee paralysis virus; ABPV, acute bee paralysis virus; BQCV, black queen cell virus; SBV, sac brood virus; KBV, Kashmir bee virus.

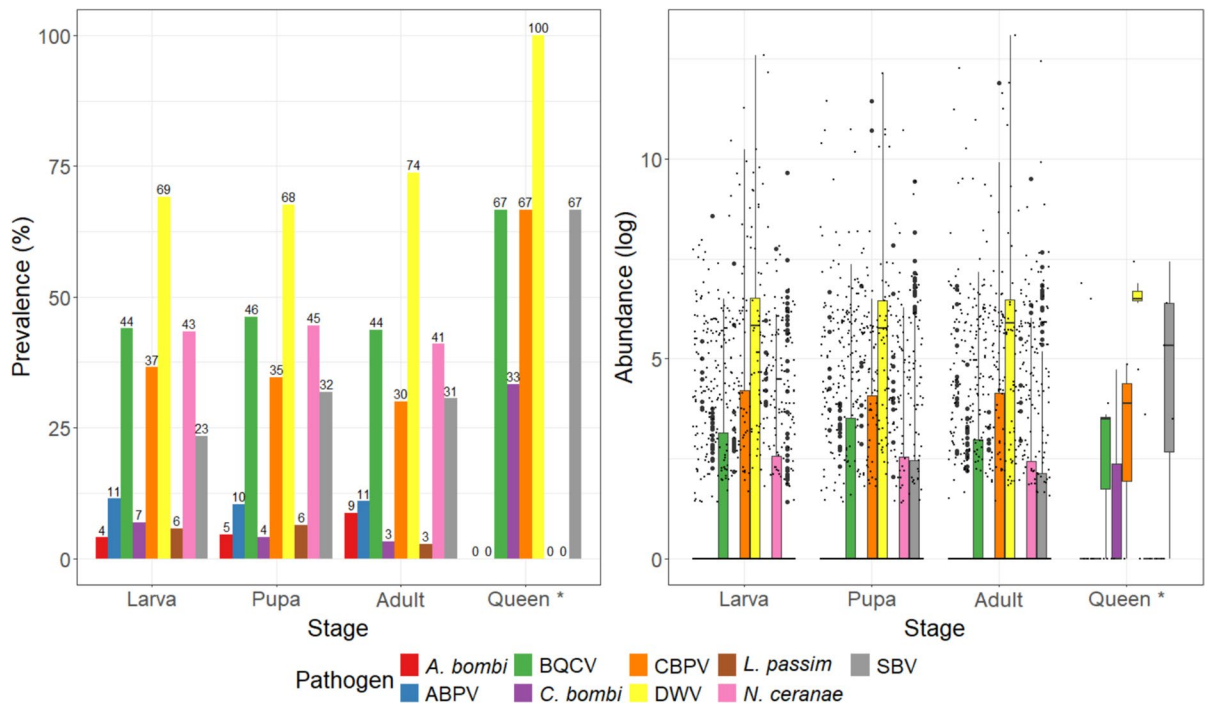


Fig. 2 Representation of each pathogen's average prevalence and mean abundance in different stages. Prevalence is indicated in percentage, and abundance is shown as a decimal logarithm. Queen* data concern queens found in three nests sampled

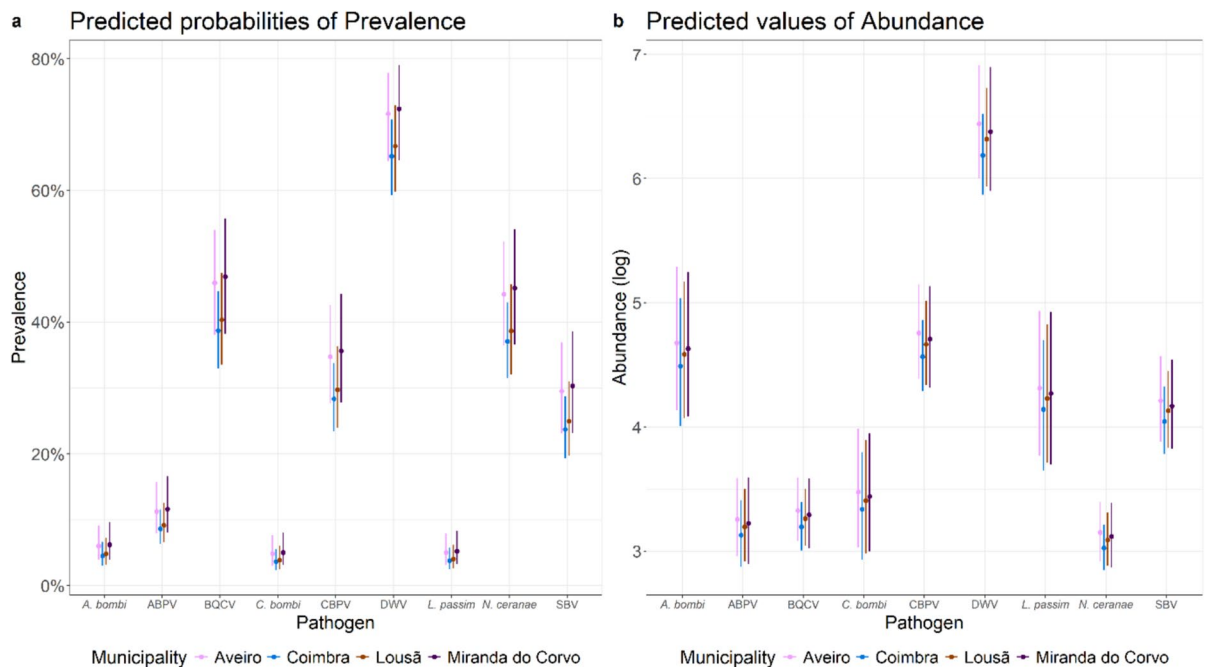


Fig. 3 Predicted probabilities of prevalence (Model AIC = 4423.3) (a) and predicted values of abundance (Model AIC = 9005.5) (b) of each pathogen in the four municipalities sam-

pled. Prevalence is indicated in percentage and abundance is shown as a decimal logarithm

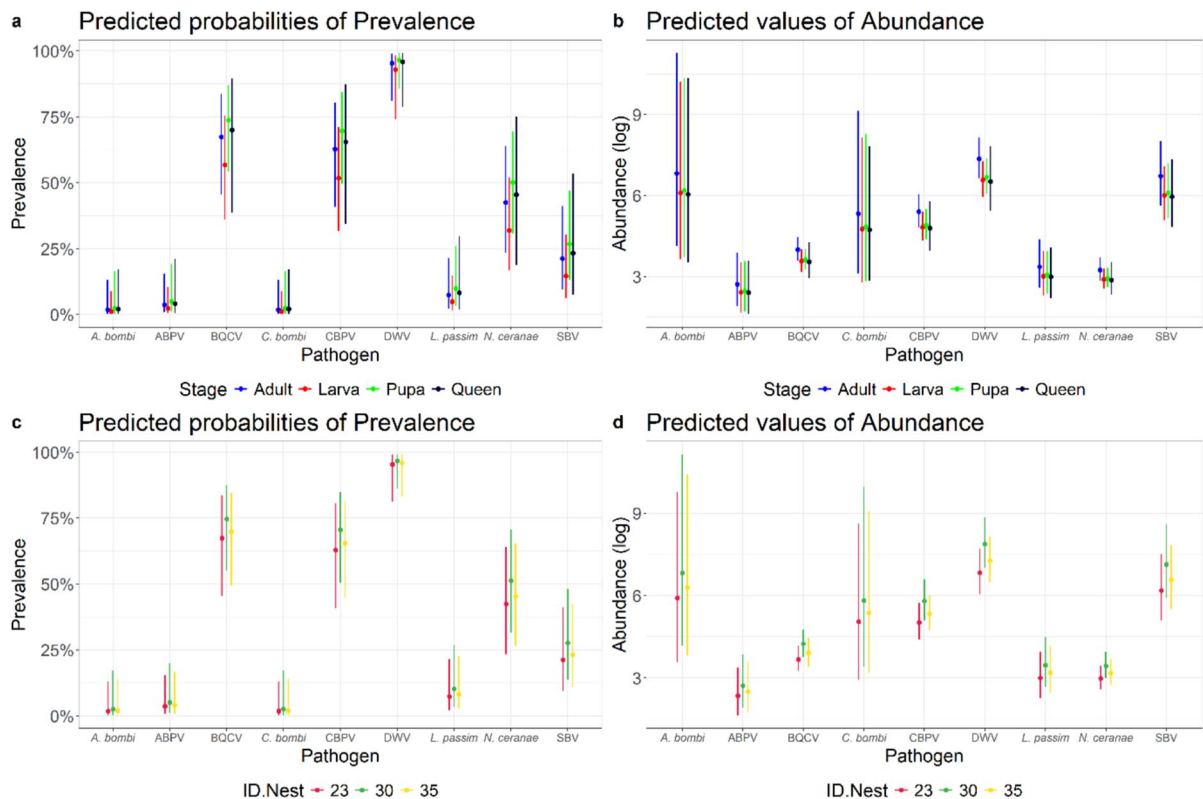


Fig. 4 Predicted probability of prevalence (Model AIC = 331.67) and predicted values of abundance (Model AIC = 816.70) of each pathogen in different stages (**a, b**) and the

three sampled nests where the queen was present (**c, d**). Prevalence is indicated in percentage, and abundance is shown as a decimal logarithm

higher occurrence of BQCV than Lousã, whereas Miranda do Corvo exhibited a higher occurrence of SBV than Coimbra. With regard to abundance, significantly higher values were observed for DWV in November compared with October, for BQCV in August compared with October and November, for SBV in August and September, and for *N. ceranae* in November compared with October (Figure S1 and Table S5).

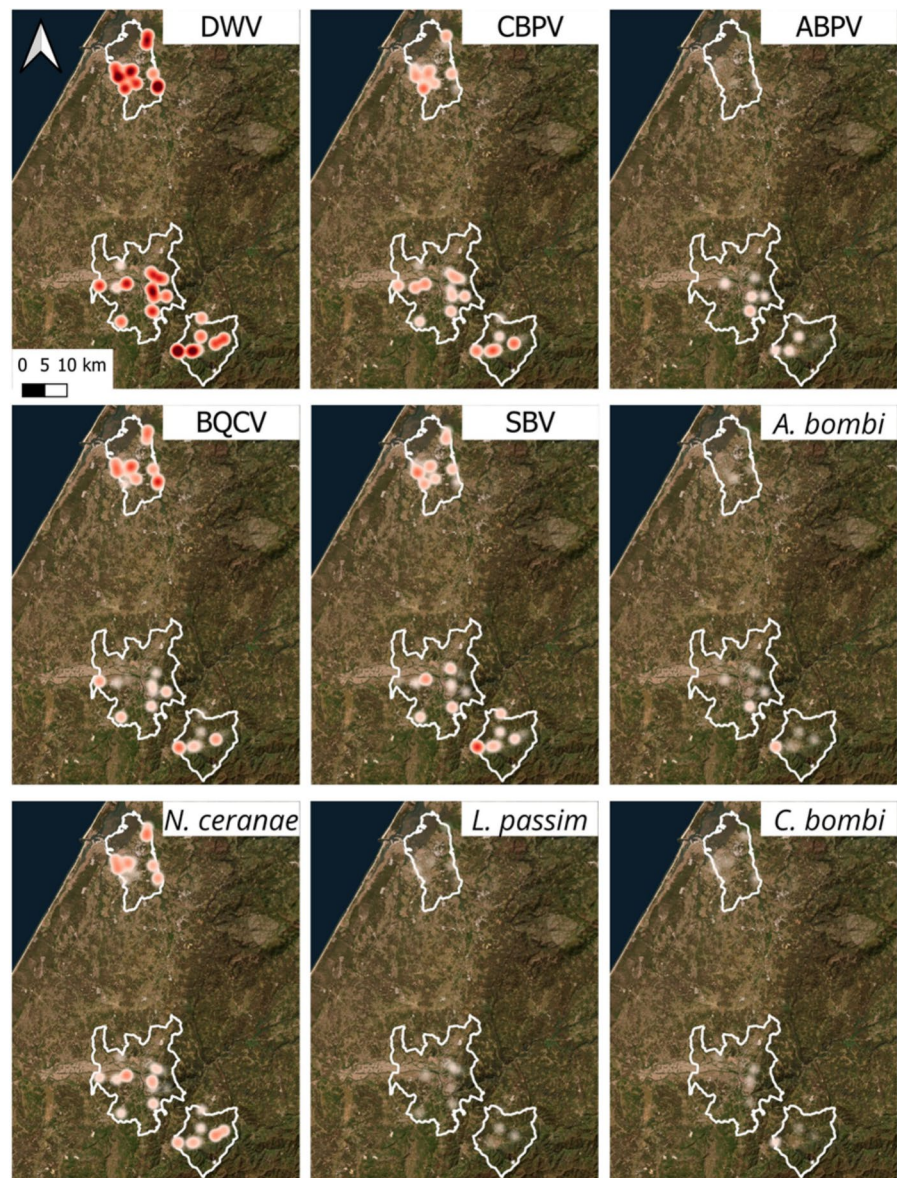
Discussion

This study identified infected *V. v. nigrithorax* individuals in nests collected in mainland Portugal, across developmental stages. These individuals were found positive for honey bee- (DWV, CBPV, ABPV, BQCV, SBV, *N. ceranae* and *L. passim*) and bumble bee-associated pathogens (*A. bombi* and *C. bombi*). Although these pathogens are prevalent

in *A. mellifera* and *Bombus* colonies (Meeus et al. 2011; Porrini et al. 2016; Bellucci et al. 2019; Bordin et al. 2022; Cilia et al. 2022c, 2024b; Plischuk et al. 2023; Tiritelli et al. 2025), they are able to infect a wide spectrum of wild bee species (Graystock et al. 2013; Dalmon et al. 2021; Cilia et al. 2022a; Tiritelli et al. 2024b) and also wasps and hornets, including *V. crabro*, *V. velutina* and *V. orientalis* (Forzan et al. 2017; Gabín-García et al. 2021; Power et al. 2022, 2023; Cilia et al. 2023; Zucca et al. 2023; Bartolomé et al. 2024; Rodríguez-Flores et al. 2024). The same pathogens are circulating in the environment affecting wild and honey bees in the Coimbra regions (Capela et al., in prep.).

To the authors' best knowledge, this is the first study to assess the prevalence and abundance of pathogens in the invasive Asian hornet in mainland Portugal. The results obtained are in line with previous findings in hornets sampled in the Spanish

Fig. 5 Descriptive heat-maps with pathogen abundance in the Portuguese municipalities of Aveiro, Coimbra, and Lousã. The maps show pathogen copy abundance expressed in log10 transformed (values range from 0 to 8) within a 2 km radius



border regions (Gabín-García et al. 2021; Bartolomé et al. 2024; Rodríguez-Flores et al. 2024).

Considering the different developmental states sampled—larvae, pupae, adults and, when available, queens—no difference was found for the presence of the pathogens detected. Interestingly, mainly DWV, BQCV, *N. ceranae* and CBPV maintained the same values in all developmental stages. These results indicate a wide spread of these pathogens within the nest, both by horizontal transmission and possible vertical transmission from the queen, as shown by the results on only 3 queens investigated. Previous

studies had mostly reported DWV presence in *V. v. nigrithorax* adults, and also in larvae, in France, Italy, and Spain (Mazzei et al. 2018; Yang et al. 2019b; Cilia et al. 2024a; Rodríguez-Flores et al. 2024), and in a few cases, specific variants like DWV-B and DWV-C were linked to individuals showing wing deformities (Dalmon et al. 2019), like just occurred for *V. crabro* (Forzan et al. 2017). The microsporidia *N. ceranae* was present in all stages of development, as previously reported in Italian and Spanish *V. v. nigrithorax* (Gabín-García et al. 2021; Cilia et al. 2024a). BQCV and CBPV are two re-emerging

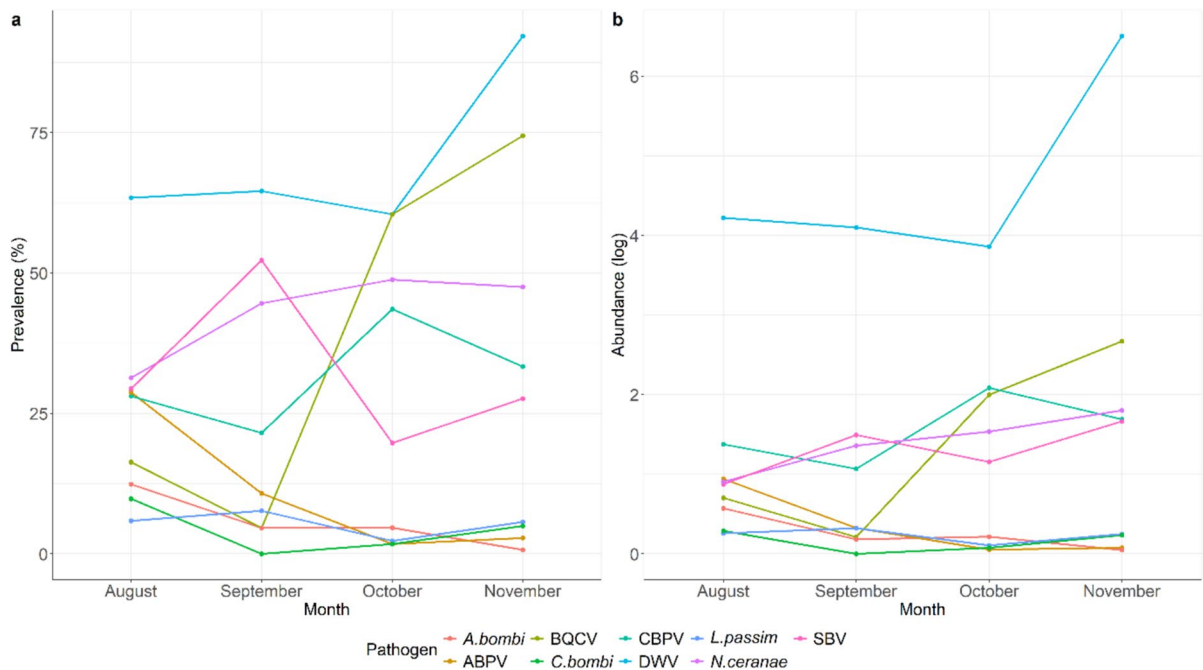


Fig. 6 Seasonal trends in the prevalence (a) and abundance (b) of each pathogen across months. Prevalence is indicated in percentage, and abundance is shown as a decimal logarithm

viruses that infect a wide spectrum of insects (Nanetti et al. 2021). BQCV has been found in some wild bee species and also in hoverflies, butterflies and mosquitoes (Bailes et al. 2018; Pislak Ocepek et al. 2022; Baril et al. 2023), while CBPV increases in incidence and prevalence during the last years (Budge et al. 2020; Cilia and Nanetti 2024; Zavatta et al. 2024).

The occurrence of *A. bombi* and *C. bombi* suggests that these pathogens are probably transmitted through predation by bumblebees as well. Indeed, the invasive hornets in its new range of expansion do not exclusively prey on honey bees but also other bees, flies, wasps, and other insects (Rome et al. 2021; Pedersen et al. 2025). The low prevalence levels of bumblebee-associated pathogens, and the high prevalence of the other pathogens investigated show that honey bees are the preferred prey of *V. v. nigrithorax*, given the ease with which they can be captured on their way out of the colonies.

Pathogen transmission in *V. v. nigrithorax* likely occurs through multiple environmental and behavioural routes. The most evident involves the ingestion of pathogens-infected bees, which are

captured and fed to larvae, as reported (Mazzei et al. 2019; Yañez et al. 2020; Power et al. 2022; Zucca et al. 2023). Also, foraging on flowers contaminated with pathogens (Burnham et al. 2021; Jones et al. 2021; Nicholls et al. 2022; Tiritelli et al. 2024a) and direct contact with infected insect hosts (Rojas-Nossa and Calviño-Cancela 2020; Rome et al. 2021; Rodríguez-Flores et al. 2022) are potential sources of infection (Garigliany et al. 2017; Yang et al. 2019a; Marzoli et al. 2020). Infections in immature stages could arise from horizontal exposure or vertical transmission. Larvae may contract pathogens directly from contaminated food or by physical contact with infected adults. Furthermore, vertical transmission from queen to egg, or venereal transmission between infected reproductive individuals, may result in the presence of infections in newly emerged adults, as observed in this study (Imhoof and Schmid-Hempel 1999; Yue et al. 2007; Mutinelli 2011; Yañez et al. 2020; Bouuaert et al. 2022).

At the intra-nest level, a study conducted in Italy found that adult individuals exhibited a higher pathogen load, likely due to their predatory behaviour (Cilia et al. 2024a). However, this conclusion was

based on data from a single nest. This discrepancy may reflect differences in the spread of the Asian hornet between the two countries: while the species is widely established across Portugal, its presence in Italy, excluding the Northwest, remains highly localized, with nests typically eliminated soon after detection in new outbreaks (www.stopvelutina.it).

The differences observed between the regions of Coimbra and Aveiro may be primarily attributed to two factors: the sampling period and environmental conditions. In Aveiro, nests were collected from October onwards, meaning they were undoubtedly secondary nests containing a high number of specimens actively foraging to feed their larvae (Diéguez-Antón et al. 2022). In contrast, some nests sampled in Coimbra at the beginning of the study were secondary nests still under development. Therefore, the sampling period could play a significant role, and the nest may function as an environmental 'bioaccumulator' of bee-associated pathogens. Moreover, the environmental characteristics of Aveiro, characterized by limited woodland and vegetation, and a predominance of coastal plains with sandy dunes, create favourable conditions for nest detection (Carvalho et al. 2020; Verdasca et al. 2021). However, these same features may also contribute to increased predatory pressure from *V. v. nigrithorax* on honey bee colonies, due to the reduced availability of alternative prey (Carvalho et al. 2020; Verdasca et al. 2021; Mata et al. 2024). Lastly, although the data show extensive nest reporting in both regions, Aveiro appears to have a higher nest density than Coimbra (<https://projects.inia.pt/gesvespa>). This could lead to an increased predatory activity on bees, potentially leading to greater pathogen transmission through predation.

Pronounced seasonal patterns were observed in both pathogen prevalence and abundance, indicating that infection dynamics in *V. v. nigrithorax* are strongly influenced by colony phenology and temporal variation in infection exposure to their prey. Both prevalence and load in November, consistent with the well-documented seasonal accumulation of pathogens in *A. mellifera* colonies during late autumn (McMenamin and Genersch 2015; Antúnez et al. 2015; Beaurepaire et al. 2020; Molinatto et al. 2025), coincide with the final phase of hornets' colony development. This period

is characterized by intense foraging and predatory activity aimed at sustaining late larval cohorts and producing reproductive individuals (Rome et al. 2015; Keeling et al. 2017). Such conditions likely enhance the intake and amplification of pathogens within nests, supporting the hypothesis that hornet nests may progressively accumulate bee-associated pathogens over the season, rather than reflecting transient or incidental exposure. Besides, season-specific peaks were also observed for other viruses, with CBPV showing higher prevalence in October, ABPV in August, and SBV in September. These heterogeneous temporal patterns likely mirror the seasonal dynamics of these pathogens in honey bee populations, where their prevalence and abundance fluctuate according to colony demography, climatic conditions, and host physiology (Grozinger and Flenniken 2019; Budge et al. 2020; Zavatta et al. 2024). The correspondence between viral peaks in hornets and known seasonal trends in bees further supports predation as a primary route of pathogen acquisition. Overall, these results indicate that *V. v. nigrithorax* nests function as temporally dynamic systems in which pathogen presence and intensity vary markedly across the season. This seasonal structuring reinforces the importance of considering sampling timing when assessing the epidemiological role of invasive vespids and suggests that late-season colonies, in particular, may represent hotspots for pathogen amplification within pollinator–pathogen networks.

From a management perspective, our findings reinforce the importance of controlling *V. v. nigrithorax* populations not only to reduce direct predation on pollinators but also to limit opportunities for pathogens circulation. The widespread presence and active replication of bee-associated viruses across developmental stages, including queens, suggests that established nests may act as focal points for pathogen amplification. Consequently, early detection and removal of nests, particularly in areas of high pollinator density (e.g., near apiaries), may provide indirect benefits for pollinators' health. While our study does not assess pathogen transmission, it provides a necessary foundation for future evaluations of disease-mediated interactions and their relevance for integrated management strategies.

Conclusion

This study provides novel insights into the presence and distribution of honey bee- and bumble bee-associated pathogens in different developmental stages of the invasive *V. v. nigrithorax*. Pathogen detection throughout the nest, including in queens, indicates widespread pathogen presence and exposure within colonies. Strand-specific RT-PCR revealed negative-strand viral RNA for several viruses, providing molecular evidence of active viral replication in *V. v. nigrithorax* and confirming that spillover from bees results in productive infection rather than passive carriage. Although the occurrence of pathogens across developmental stages is consistent with multiple potential exposure routes within nests, the relative contribution of horizontal and vertical transmission could not be assessed. Likewise, while viral replication suggests that *V. v. nigrithorax* is unlikely to represent a strict dead-end host, transmission to other species was not assessed. Consequently, this species can be considered a biologically permissive spillover host for several bee-associated viruses, whereas its potential role as a spillback reservoir remains to be proved. Moreover, the widespread infection by these pathogens opens new avenues for the exploration of targeted biocontrol strategies, although targeted studies are needed to evaluate the fitness effects of these infections on *V. v. nigrithorax*. Continuous surveillance and integrated pest management strategies remain essential to mitigate the threat posed by this invasive species.

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Author contributions NC and GC contributed to the study conception and design. Sample collection was performed by SS, NC, PS and GC, while laboratory analysis were performed by RT, VC, and GC. Data analyses were performed by RT and GC. The first draft of the manuscript was written mainly by SS and GC, while all authors commented on previous versions

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Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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